

a4a preliminary runs

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Introduction

The a4a method utilizes catch-at-age data to derive estimates of historical population size and fishing mortality. However, unlike XSA, model parameters estimated using catch-at-age analysis are done so by working forward in time and analyses do not require the assumption that removals from the fishery are known without error. The stock assessment model framework is a non-linear catch-at-age model implemented in R and FLR, and using ADMB that can be applied rapidly to a wide range of situations with low parametrization requirements. The model structure is defined by submodels, which are the different parts that require structural assumptions. There are 5 submodels in operation:

a model for F-at-age

a model for the initial age structure

a model for recruitment

a (list) of model(s) for abundance/biomass indices catchability-at-age

and a list of models for the observation variance of catch-at-age and abundance/biomass indices.

The submodels form use linear models. This opens the possibility of using the linear modelling tools available in R: see for example the mgcv (<http://cran.r-project.org/web/packages/mgcv/index.html>) gam formulas, or factorial design formulas using.

Materials and Methods

The statistical catch-at-age was conducted using using the FLA4a package in R (part of the FLR-project, Kell et al., 2007; <http://www.flr-project.org/>).

[...]

The available CAS data covered the period 1985-2018 and the catch at age (CAA) data were generated using a deterministic age slicing. The available CAS did not justify splitting CAS into ages greater than 6 and in line with the previous assessment XSA runs were conducted with a plus group of 6. As in 2016 continuity assessment, [...] natural mortality was assumed to equal 0.2 for all ages and years. In the 2016 assessment it was assumed that fish first mature at age 3 (when 50% are mature) and are fully mature at older ages. Considering new information (ICCAT 2020), it was currently assumed 15% and 65% maturation at ages 2 and 3 respectively. Full maturity was considered from age 4 onwards. Weights-at-age were derived from the mixture analysis and were consistent with the CAA. Figures 1 and 2 illustrate the available CAA data by age and year.

Five standardized CPUE series were considered: Greek longliners (SCRS 2020/021), Moroccan longliners (SCRS 2020/026), Spanish longliners (SCRS 2020/043), Ligurian longliners (...) and Sicilian longliners (...). The standardized CPUE biomass indices were considered to be representative of the 1-3 age-group abundances. Fleet catchability was assumed to be independent of year-class size for all terminal years.

Data

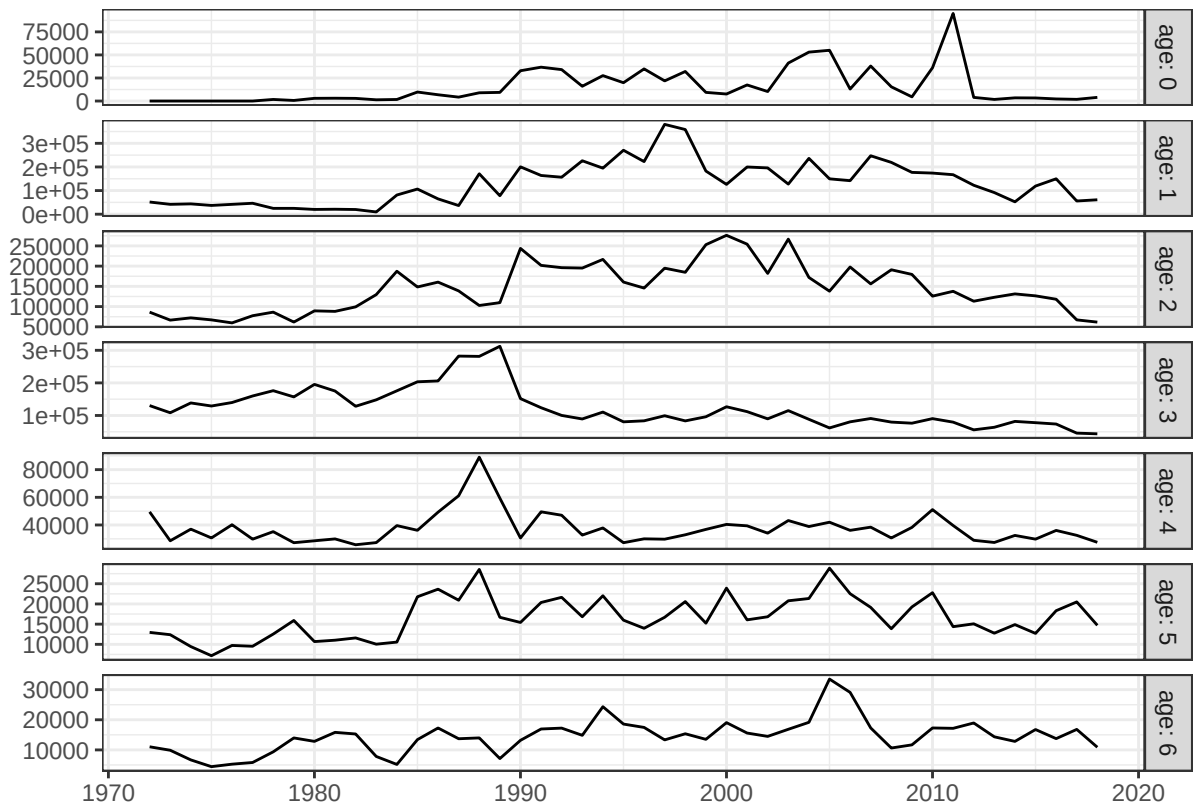


Figure 1. Time series of catch-at-age.

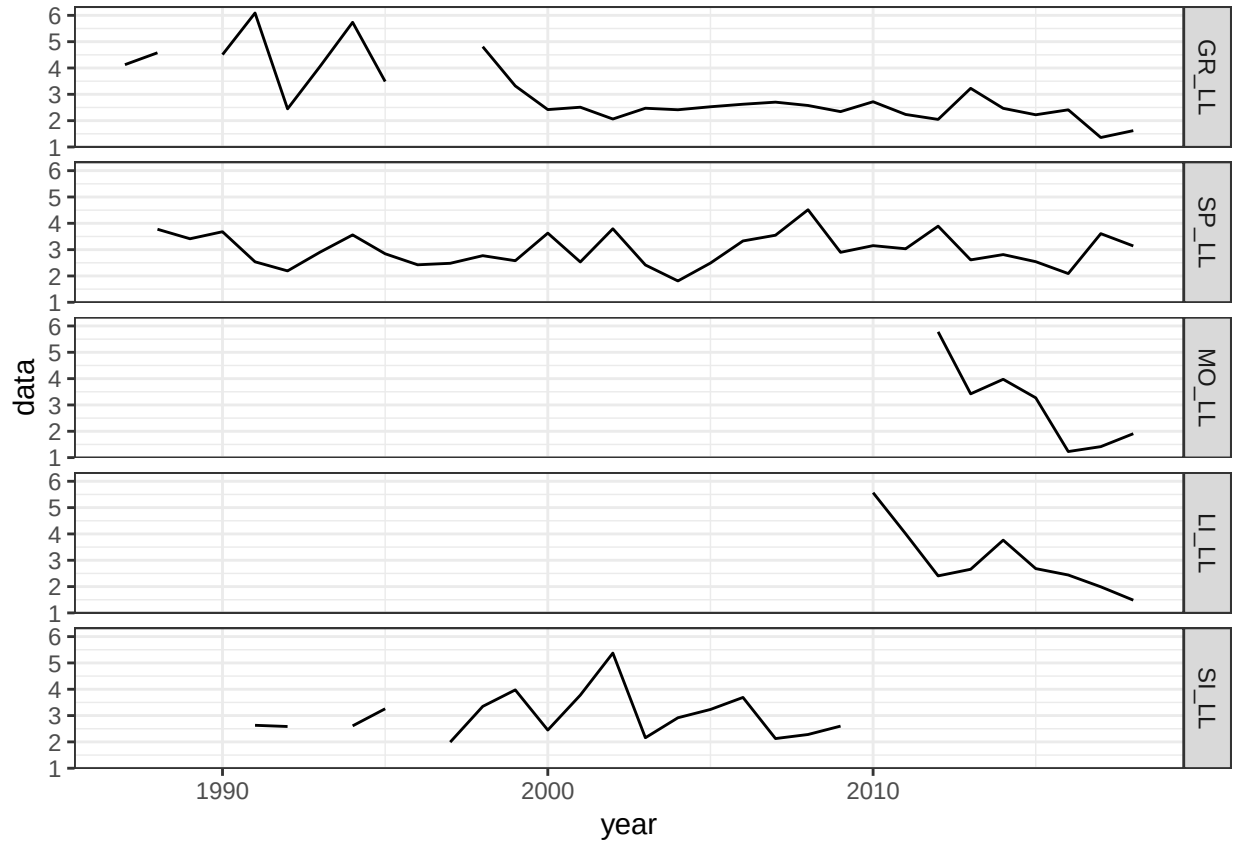


Figure 2. Time series of biomass CPUEs.

1st trial

a4a submodel setup

fishing mortality at age

$$F \sim s(year, k = 20) + s(age, k = 5)$$

recruitment model

$$SR \sim factor(age)$$

variance

Assumed constant for the indices and effect on a age $\sim s(age, k = 3)$ for catch observations.

Results

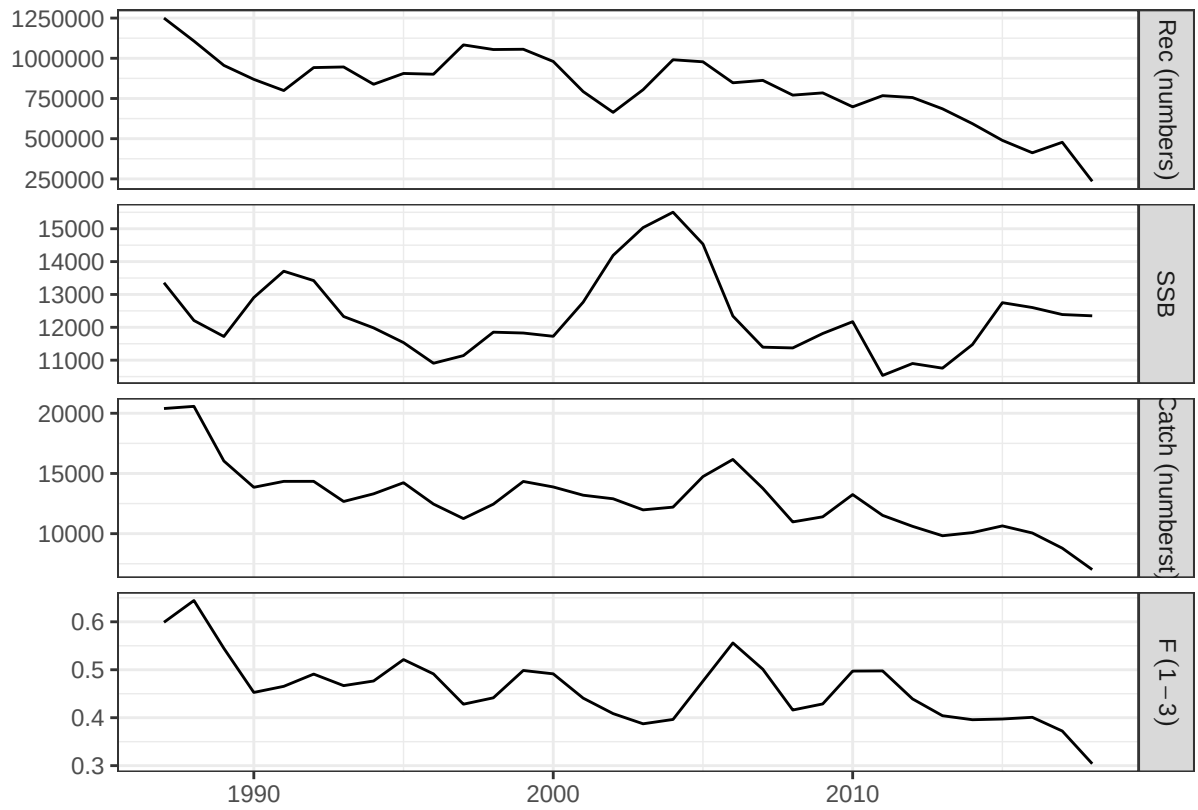


Figure 3. Assessment Results.

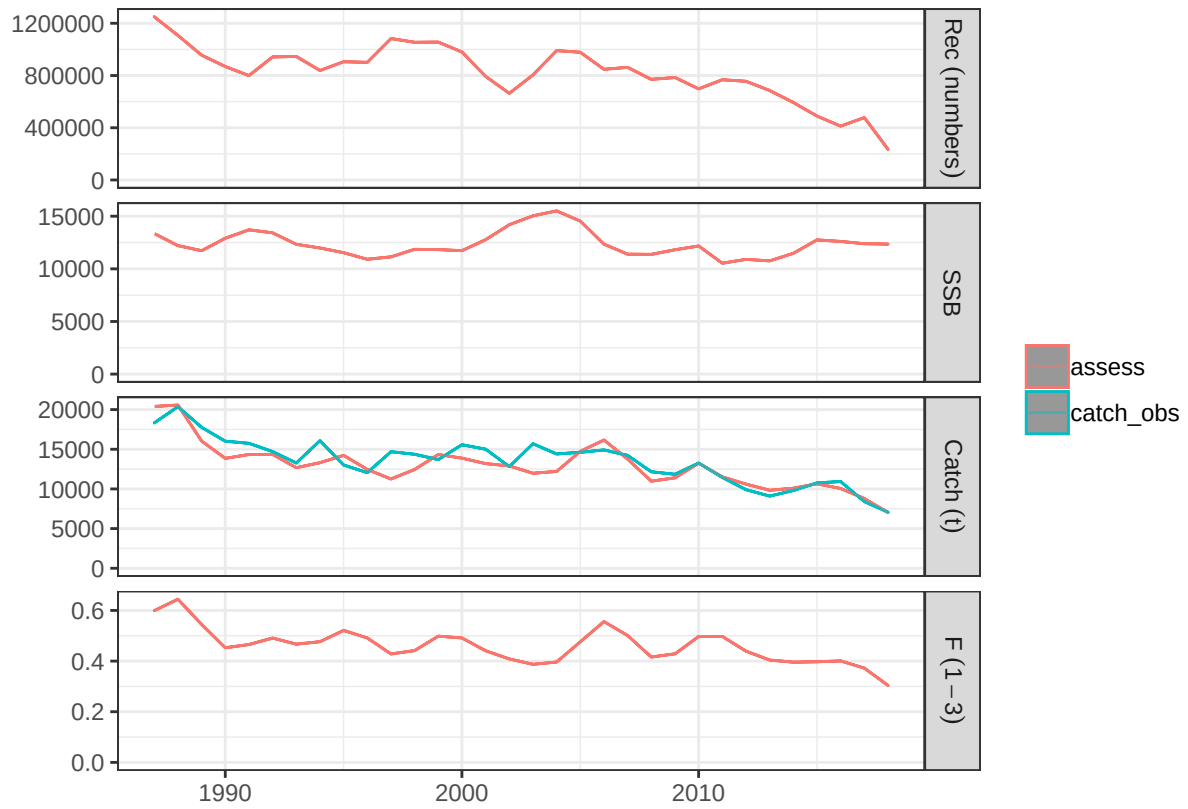


Figure 4. Assessment Results against observed catch.

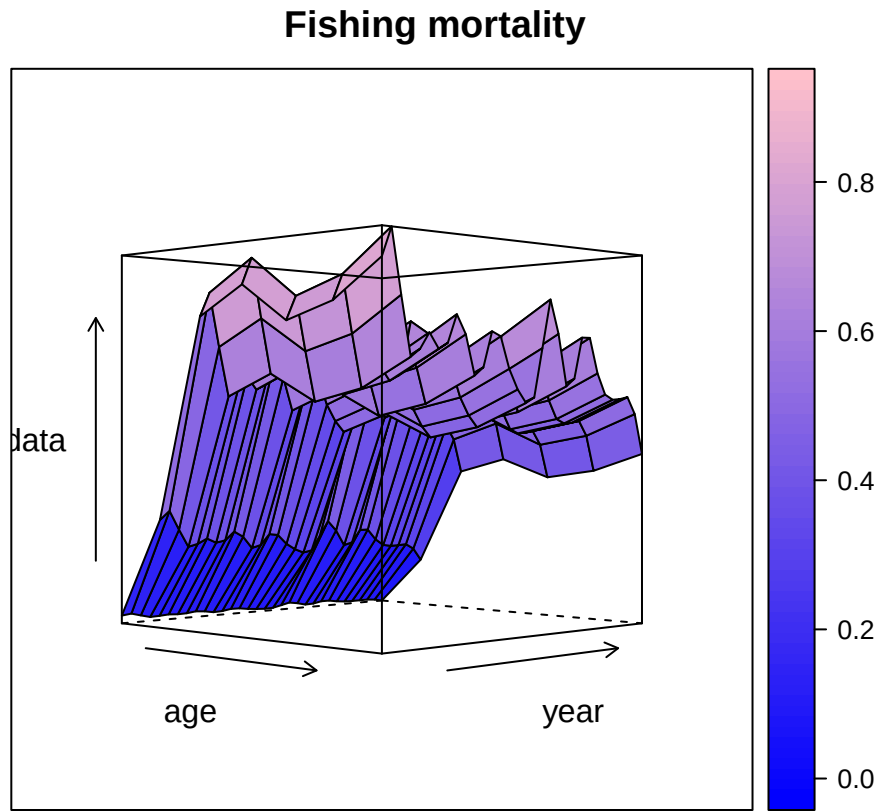


Figure 5. Fishing mortality 3D plot.

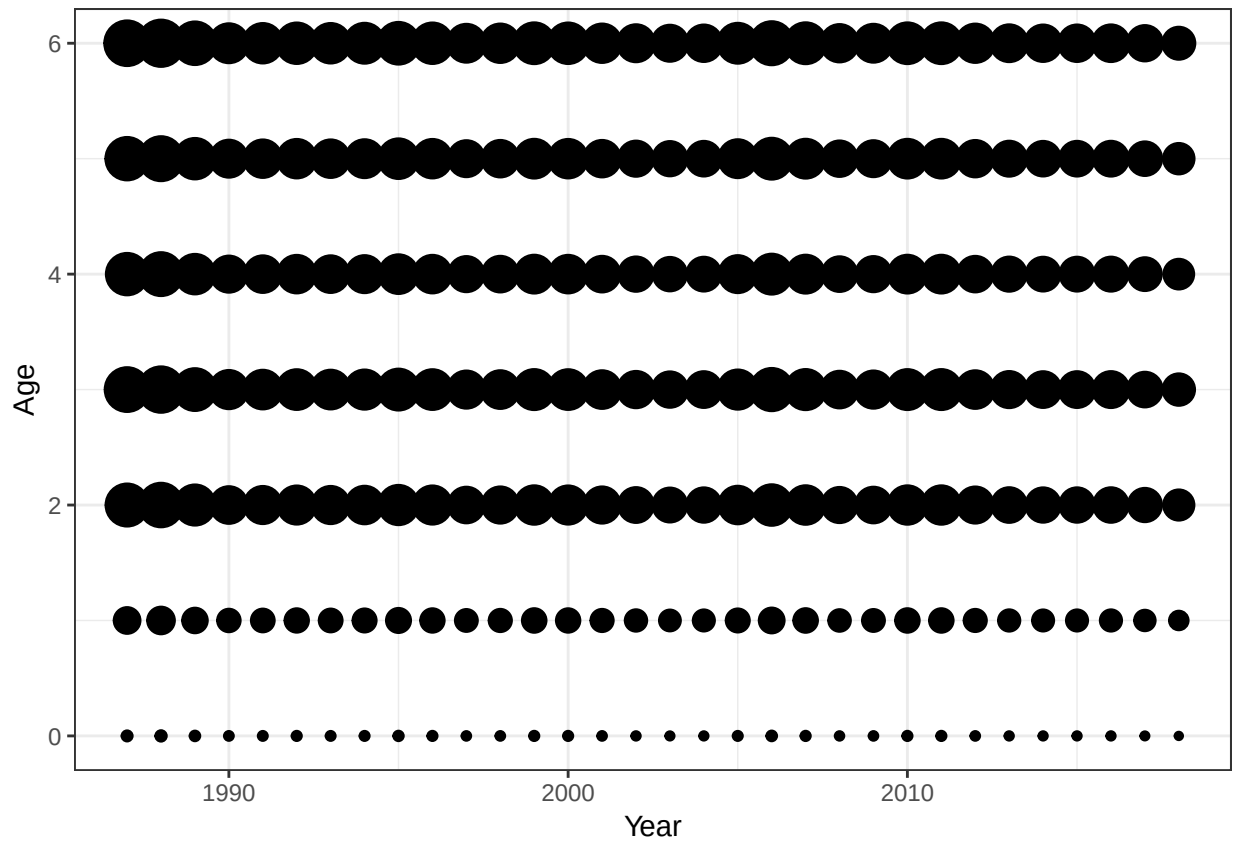


Figure 6. Fishing mortality bubble plot.

Diagnostics

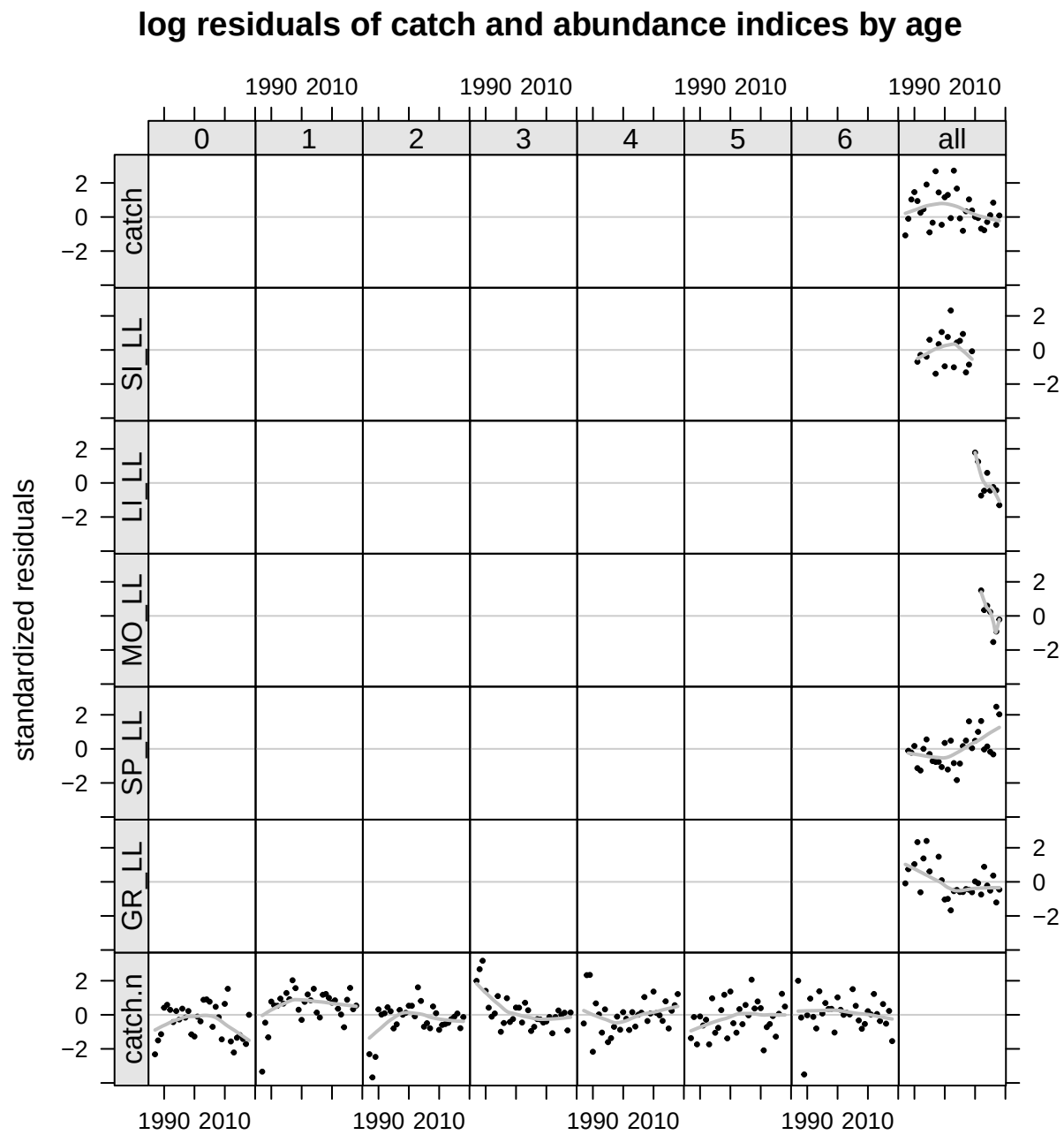
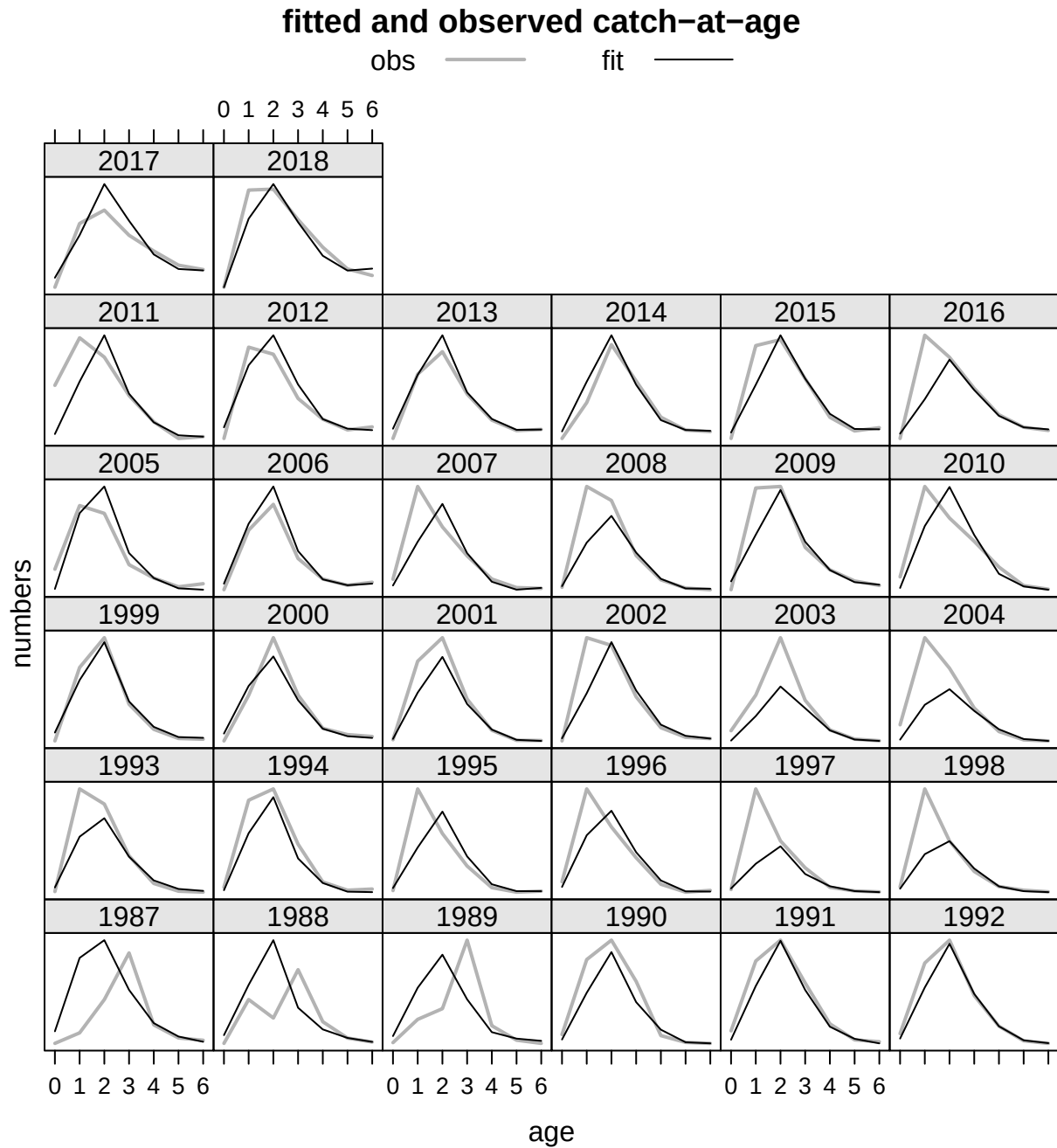


Figure 7. Log residuals against fitted values.



2nd trial

Remove old index and smooth a little the f model

a4a submodel setup

```
idx_bio2 = FLIndices(idx_bio[1],idx_bio[2],idx_bio[3],idx_bio[4])
```

fishing mortality at age

$$F \sim s(year, k = 15) + s(age, k = 5)$$

recruitment model

$$SR \sim factor(age)$$

variance

Assumed constant for the indices and effect on a age $\sim s(age, k = 3)$ for catch observations.

Results

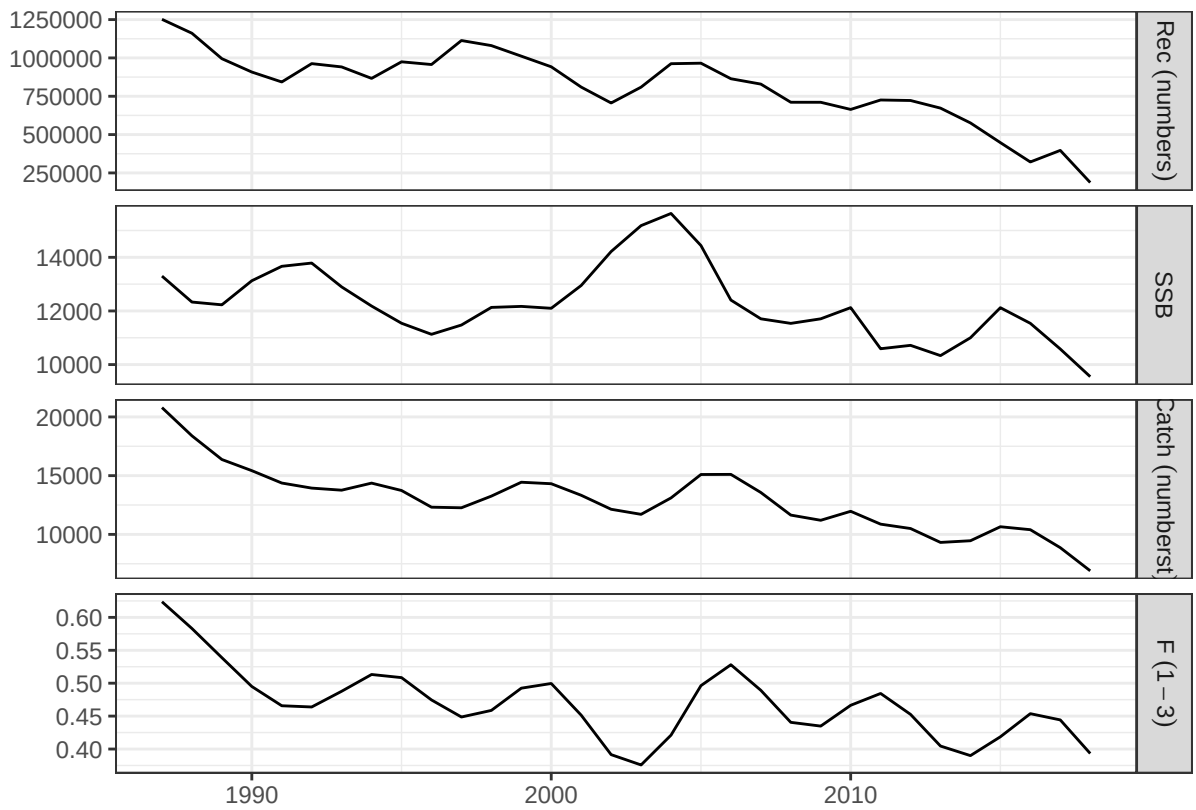


Figure 8. Assessment Results 2nd run.

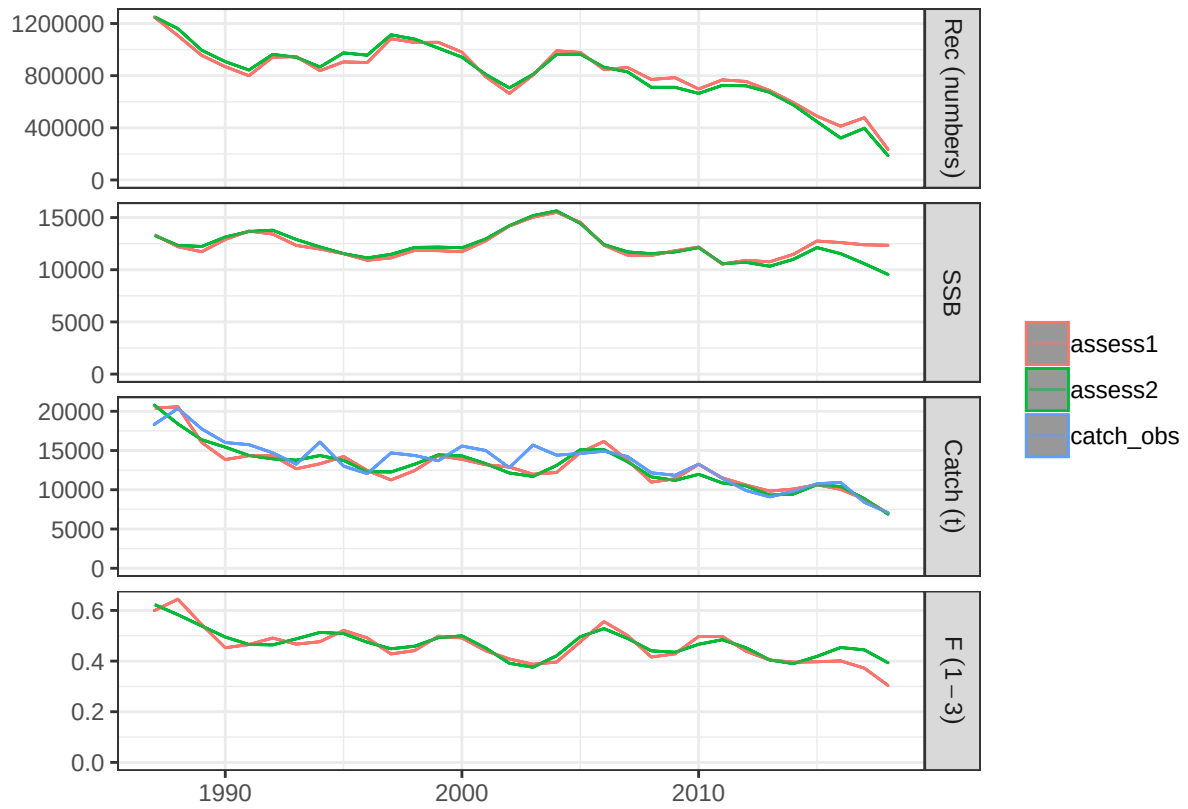


Figure 9. Assessment Results against observed catch and 1st run.

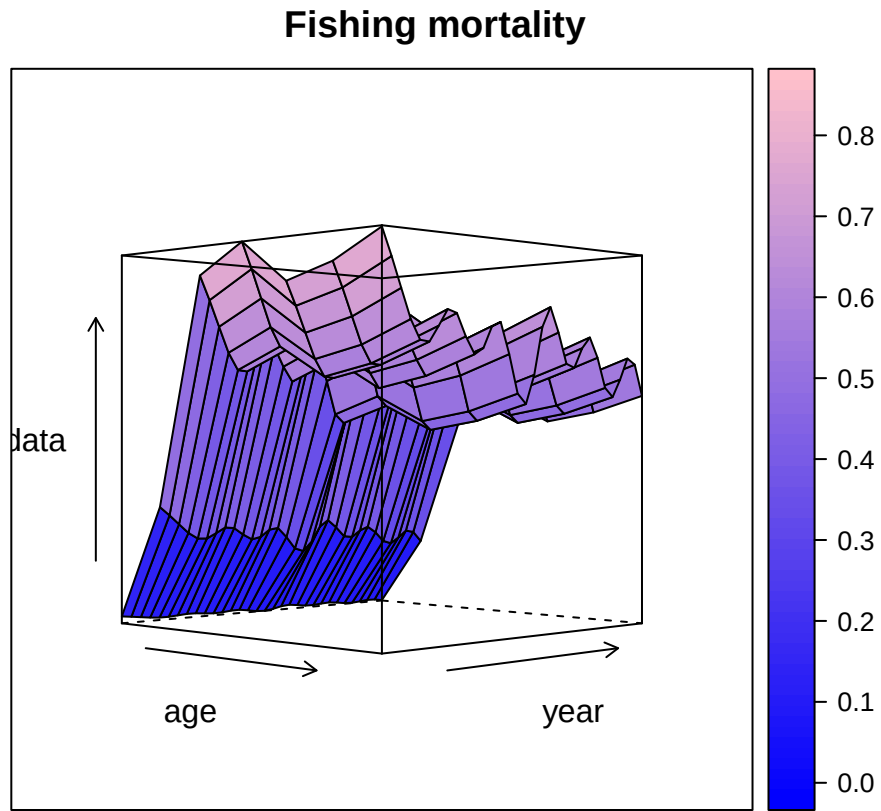


Figure 10. Fishing mortality 3D plot.

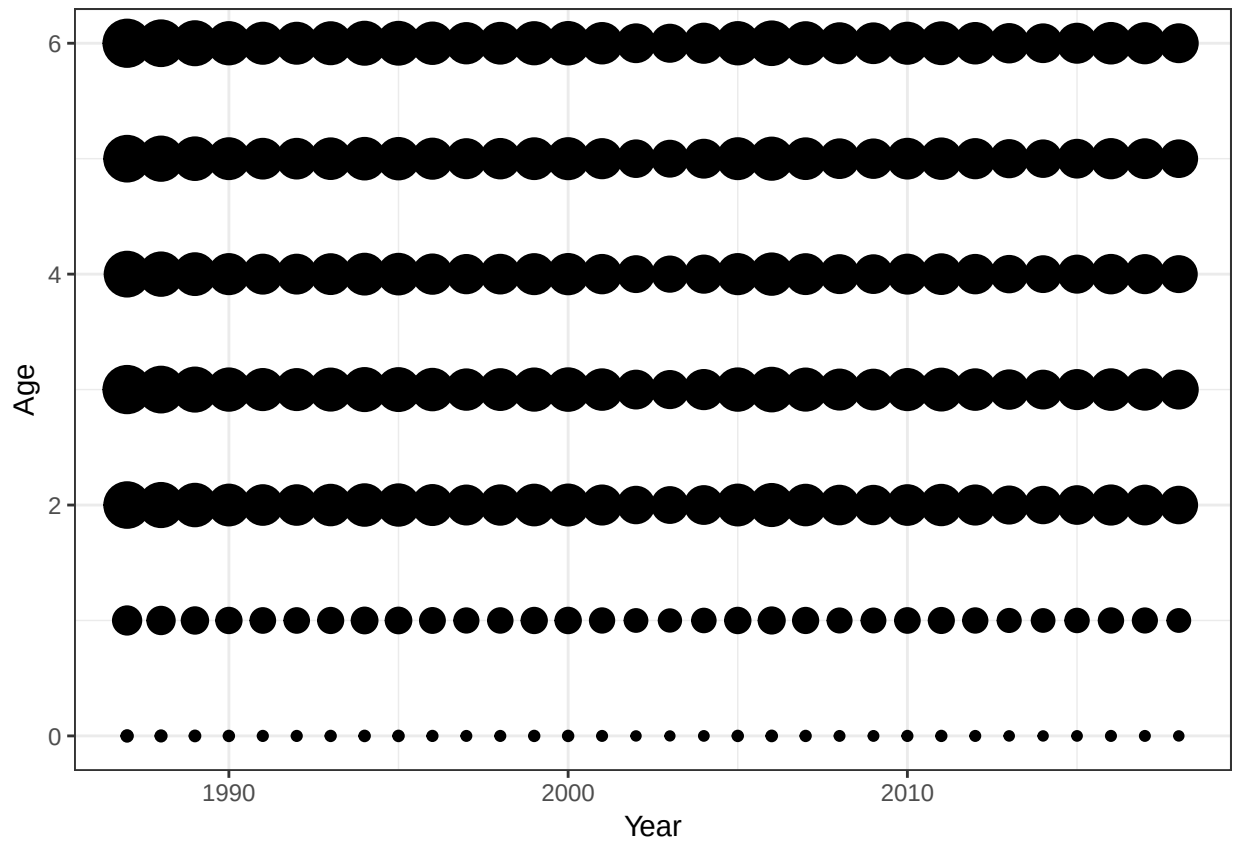


Figure 11. Fishing mortality bubble plot.

Diagnostics

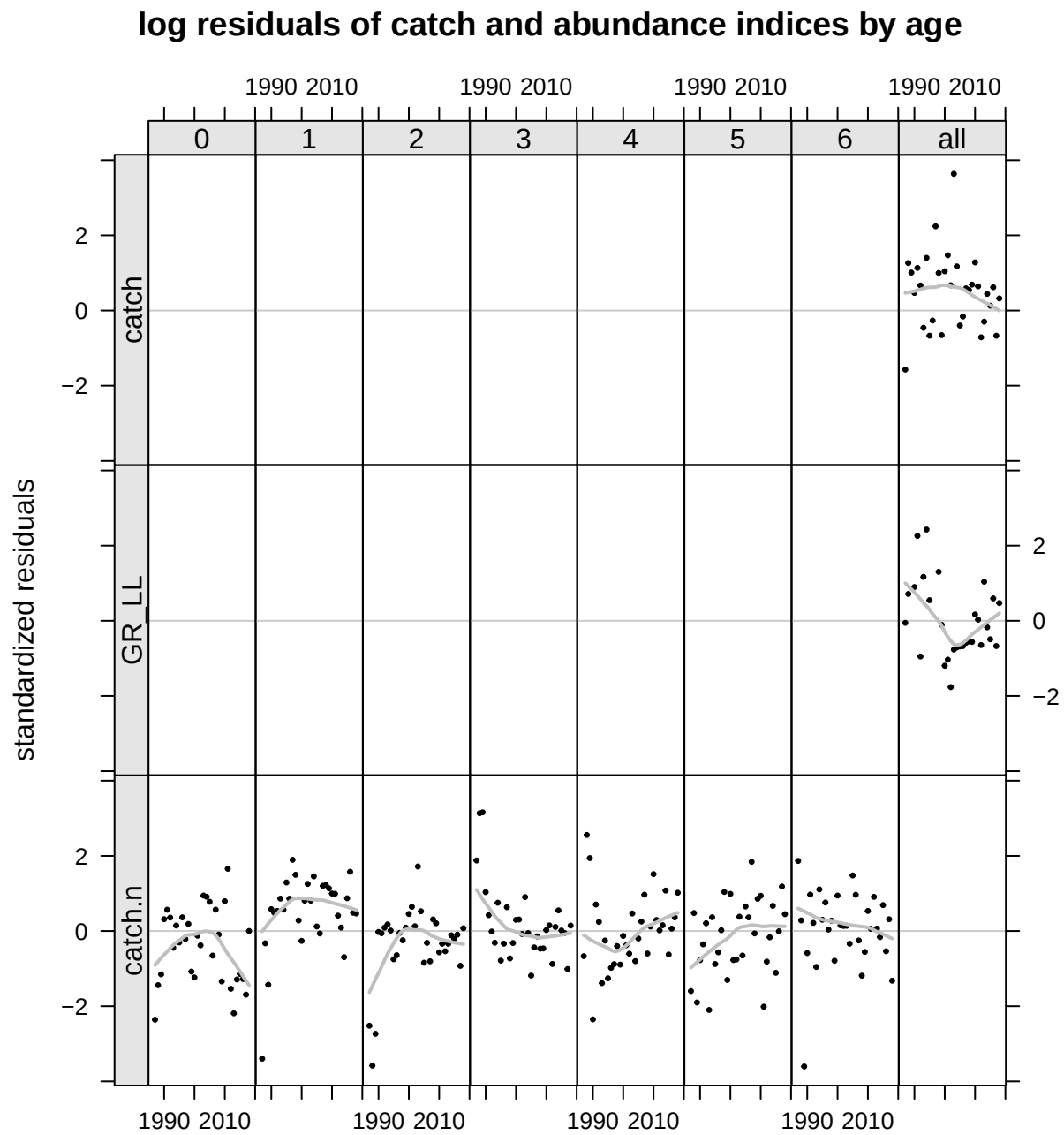


Figure 12. Log residuals against fitted values.

